Institutional Ownership, Blockholders, and Attention to the Future*

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Abstract

Corporate short-termism is, at its root, an agency problem: shareholders would have managers look ahead, yet the work is costly, difficult to contract upon, and rewarded only later, so managers may undertake too little of it. How far ahead they in fact look has rarely been observed. We construct a direct firm-level measure of it from the horizon of forward-looking statements in earnings calls: the median number of months ahead to which a firm's managers refer, which we term its *attention to the future*. The measure predicts subsequent investment, R&D, and innovation, and it contracts as a firm approaches financial distress. We then study how corporate governance shapes this attention, developing a model of managerial attention that nests price discipline through informed trading [Edmans and Manso, 2011] and the control–initiative tradeoff under blockholder intervention [Burkart et al., 1997]. Both predictions hold in the data: within firms, attention rises with the level and breadth of institutional ownership and falls as ownership concentrates in a single block. Consistent with the model's intervention mechanism, the block's negative association weakens as managerial equity rises, and liquidity-based exit alternatives do not account for it. Ownership thus shapes not only what firms ultimately do, but how far ahead their managers look before acting.

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1 Introduction

A large literature in corporate finance asks how shareholders induce managers to take costly actions on the firm's behalf. One of the most consequential of those actions is preparing the firm for the future. Before a firm can invest, innovate, or adapt, its management must work out what lies ahead: which opportunities are coming, what distant consequences today's choices carry, and how to prepare for them. That work is costly today, its return is distant, and it is difficult to contract on. Economists have long argued that managers may undersupply it and sacrifice long-run value to near-term performance [Stein, 1989, Graham et al., 2005, Edmans, 2009, Asker et al., 2015, Terry, 2023]. Short-termism is thus an agency problem in the *temporal* allocation of managerial attention: the concern is not that managers work too little, but that they do not look far enough ahead. Yet the object at the center of that claim, how far ahead management in fact looks, has rarely been measured in a cardinal unit.

The literature has largely observed the residue of looking ahead rather than the act itself. R&D, investment, and patents record what a firm *did*, not how far ahead it *looked*, and they move for many reasons besides temporal orientation; the inference back from the residue is not even signed, since looking only a short way ahead can produce underinvestment and overinvestment alike [Bebchuk and Stole, 1993]. We construct a direct measure of *revealed attention to the future* from corporate language. We define it as how far into the future management searches for opportunities, evaluates their consequences, and plans for them, and we construct a firm-year measure from the earnings-call transcripts of U.S. public firms: the median number of months ahead referenced in the firm's forward-looking statements. The result is a cardinal measure, in months, of how far ahead a firm looks. It reads what management articulates, not the full plan it may hold internally: it is a noisy but informative proxy for how far ahead the firm looks, not a transcript of its cognition. We do not claim a longer horizon is always optimal, only that it is a value-relevant input into long-run action, and we show below that it predicts what firms subsequently do.

Like any costly action the manager alone can supply, how far ahead she looks is shaped by who owns the firm. We study it through two classic ownership mechanisms, both originally written about effort or initiative and revisited here for the manager's horizon. The first is discipline through informative prices [Holmström and Tirole, 1993]: the firm's informed owners trade on its long-run value and impound it into the price, rewarding an equity-holding manager today for work whose payoff lies ahead. Edmans [2009] applies this logic to managerial myopia, and Edmans and Manso [2011] shows that competition among informed owners makes ownership *breadth* a margin in its own right. The second mechanism is the

control–initiative tradeoff [Aghion and Tirole, 1997]. In Burkart et al. [1997], anticipated intervention by a concentrated owner substitutes the block's judgment for the manager's and deters her noncontractible initiative. The two mechanisms thus concern the two kinds of owner that dominate the modern firm: institutional investors, whose informed trading shapes the price, and a concentrated blockholder, who can intervene directly. Both are pervasive today. Institutions hold about 80% of the S&P 500 by 2017, up from under 40% in 1980 [Gompers and Metrick, 2001, Backus et al., 2021], while roughly 96% of public firms have at least one blockholder [Holderness, 2009]. Modern ownership is thus both institutional and concentrated, making price discipline and intervention simultaneously relevant.[1] Does such ownership make the firm look further ahead, or nearer?

Neither mechanism was written about time: in both, the manager chooses effort or initiative, not how far ahead to look. We apply both mechanisms to a model of how far ahead the manager chooses to look, in which a controlling blockholder emerges from a fixed cost of control [Shleifer and Vishny, 1986]. The model yields two families of predictions. The first concerns informed ownership: greater breadth makes the price more informative and extends how far ahead she looks; once a larger institutional ownership stake also sharpens signal precision, total institutional ownership works the same way. The second concerns intervention. A larger block shortens it when the private benefit of retaining control exceeds the manager's equity-linked return from a better project. A larger managerial stake attenuates this deterrent: a negative concentration effect should weaken as insider ownership rises, the prediction we take to the data. These implications organize the empirical analysis.

Before relating the measure to ownership, we establish that it captures economically meaningful behavior. The typical forward-looking statement points about five months ahead; over half of the measure's variation is within firm, and it is persistent from year to year. The measure passes two tests. First, it predicts real forward-looking activity. A one-standard-deviation increase is followed a year later by capital expenditure higher by about 0.25 percentage points of assets and R&D higher by about 0.8 percentage points (roughly 6 and 9 percent of their sample means), and by faster patent growth and more valuable, more cited patents [Kogan et al., 2017]. The investment result survives controls for the firm's cost of capital and perceived discount rate [Gormsen and Huber, 2025]. Second, we show that the firm's circumstances shape its attention to the future. Firms in trouble pay less attention to the future, consistent with the underinvestment that debt overhang predicts for distressed firms [Myers, 1977]. Within a firm, attention contracts after an earnings miss, a stock-price

[1]The papers do not divide cleanly: Edmans and Manso [2011] model trading and intervention, and a large owner may also sharpen the price through informed trading [Edmans, 2009, Admati and Pfleiderer, 2009]. We use them as representatives of the two mechanisms.

crash, an operating loss, high leverage, or elevated risk, and expands with cash and distance to default. The measure therefore co-moves systematically with financial distress.

We relate the measure to ownership in a 2002–2020 panel of 35,416 firm-years. Our ownership data come from the Ownership Project [Amel-Zadeh et al., 2022], which reconstructs institutional investors, blockholders, and insiders in a single consistent panel; this integrated coverage is what lets us measure all three margins jointly, and later inside equity alignment, on the same firms. The 13F institutional ownership share and the number of institutional owners proxy for the level and breadth of institutional ownership; the largest institutional owner's share proxies for concentration. We enter all three measures jointly, absorbing firm and sector×year fixed effects, so each margin is measured holding the others fixed and no comparison across firms enters. Coefficients are reported as a percentage of the dependent variable's within-firm standard deviation (one such standard deviation is 1.35 months of attention). A one-standard-deviation increase in the level is associated with about 7% of a within-firm standard deviation of additional attention; a one-standard-deviation increase in breadth with 11–15%, the largest of the three margins; and a one-standard-deviation increase in concentration with about 3.7% *fewer*, a contraction comparable to the one a firm shows after missing earnings. These are not large numbers in absolute terms, but they are of the same order as what financial distress does to the measure: a one-standard-deviation change in leverage or distance to default moves attention about as much. The concentration fact is not an artifact of one definition: it holds for the gap to the second-largest owner, an ownership Herfindahl, blockholder presence, and the share held by large owners alike.

A natural concern is the reverse reading: forward-looking firms may attract particular owners, rather than ownership shaping how far ahead managers look. We use Russell index assignment as a separate test of the direction of causality. The annual reconstitution of the Russell 1000 and 2000 indexes provides quasi-random variation in the index-driven passive share: it moves index-tracking money across otherwise similar firms at the cutoff [Appel et al., 2016]. Crossing into the Russell 2000 raises passive ownership by about 1.1 percentage points (first-stage $F = 103$). A one-percentage-point increase in passive ownership raises attention by 0.17 months at $t{+}1$, while the pre-period estimate is statistically indistinguishable from zero. Because index assignment, rather than the firm's prior language, determines the passive inflow, this result shows that index-induced passive ownership can move attention and weighs against an explanation based solely on forward-looking firms attracting owners.[2]

[2]We read this as evidence that institutional ownership *causally* extends attention, not as a validation of any single channel. Our framework formalizes the price channel for tractability, but an exogenous ownership inflow can also lengthen the manager's horizon through the voice and monitoring that index investors are documented to exert [Appel et al., 2016, 2019] and through the greater information production that institutional ownership encourages [Boone and White, 2015]. Because index funds are unlikely to be the informed

We next test the model's prediction that inside equity alignment attenuates the block's deterrent effect. The same Ownership Project panel now lets us bring insiders alongside institutions and blockholders. We use insider ownership reported on Forms 3–5 as a broad measure of inside equity alignment. Officers include the executives who choose and communicate firm policy, while directors and 10-percent beneficial owners extend the measure to firm-level equity held by statutory insiders. The resulting measure is well suited to testing the model's mechanism: whether inside equity alignment moderates the effect of a large block. Its interaction with the largest institutional owner's share is positive, statistically significant, and stable as controls enter (1.7% to 1.4% of a within-firm standard deviation per unit of the product of the standardized ownership variables), while the block's average effect stays negative (-4.6% to -5.7%). The moderation is economically comparable to the block effect itself: a one-standard-deviation increase in insider ownership shrinks the block's deterrent by about 30%, and over a two-standard-deviation swing it nearly halves. Across the observed range, inside equity attenuates the block's deterrent while its average effect remains negative.

The natural alternative reading is exit threat rather than intervention: in exit models, too, governance transmits through the manager's price-linked wealth, and empirically the blockholder's effect on value and reporting has been shown to strengthen with managerial price sensitivity [Bharath et al., 2013, Dou et al., 2018]. The evidence is inconsistent with a simple liquidity-based exit account. The exit channel's own proxies (stock liquidity and share turnover, each interacted with insider ownership) either have no effect or enter with the wrong sign, and neither absorbs the concentration-by-equity interaction. It is likewise robust to absorbing nonlinearity in the block and to residualizing insider ownership with respect to block size. Together, the contrasting effects of ownership level, breadth, and concentration, and the moderation by insider ownership, align with the model's two mechanisms.

The paper proceeds as follows. Section 2 relates the paper to the closest literature. Section 3 develops the conceptual framework and its predictions. Section 4 builds and validates the measure. Section 5 presents the baseline ownership evidence and the Russell design. Section 6 tests the concentration–equity prediction and evaluates the alternative exit mechanism. Section 7 concludes.

traders of the price channel, the effect plausibly runs through these governance channels; either way, the reduced-form causal estimate does not depend on which operates. Cleanly separating the channels would require a further instrument that isolates one of them, and credible instruments for ownership are scarce [Edmans and Holderness, 2017]; we therefore do not attempt to.

2 Related Literature

This paper connects three literatures: text-based measurement of managerial attention, informed ownership, and blockholder control. We measure firms' attention to the future from the reach, in months, of their forward-looking language. This common outcome lets us study institutional ownership along three distinct margins: level, breadth, and concentration. Price discipline organizes the first two; the control–initiative tradeoff yields the prediction that inside equity moderates the effect of concentration.

Measuring attention to the future. Our use of corporate text follows the broader text-as-data literature [Gentzkow et al., 2019, Hassan et al., 2025]. A closely related strand constructs firm-level measures of political, epidemic, and country risk from earnings calls [Hassan et al., 2019, 2023, 2024]. Closest in object, Song and Stern [2024] use 10-K filings to measure how much attention firms devote to macroeconomic news, not how far into the future that attention reaches. In accounting, other measures of corporate horizons have been proposed: Brochet et al. [2015] compare short- and long-horizon language in conference calls. Related work codes the forward reach of call language [DesJardine and Bansal, 2019] or collects stated project durations from regulatory filings [Dessaint et al., 2023]. Relative to these measures, ours provides a scalable firm-year measure, in months, of how far ahead management's language reaches. We interpret it as *revealed attention in language*, not direct observation of cognition.

Institutional ownership and price discipline. Price discipline through informed trading links institutional ownership to attention to the future. Following Holmström and Tirole [1993], a price that aggregates information about long-run value rewards an equity-holding manager for actions whose payoffs lie ahead. Edmans [2009] applies this logic to myopia, showing how informed blockholder trading can encourage long-run investment, and Edmans and Manso [2011] makes ownership breadth central: competition among informed owners impounds more information into price. We apply this incentive force to a choice with an explicit temporal reach and allow both the stock of informed institutional capital and the number of informed owners to affect informativeness. The model therefore speaks separately to institutional ownership level and breadth.

Empirical work reaches different conclusions depending on investor horizon: transient or high-churn institutions are associated with more myopic policy [Bushee, 1998, 2001, Hotchkiss and Strickland, 2003, Gaspar et al., 2005, Yan and Zhang, 2009], whereas longer-horizon owners are associated with innovation and investment [Wahal and McConnell, 2000,

Aghion et al., 2013, Derrien et al., 2013, Bena et al., 2017, Harford et al., 2018].[3] Rather than proxy firm horizons with accounting-based policies or realized innovation outcomes after classifying investors as patient or transient, we place the firm's own temporal reach on the left-hand side and estimate the level and breadth margins jointly.

Blockholder control, managerial initiative, and exit. The concentration mechanism begins with the distinction between formal and real authority in Aghion and Tirole [1997]: an agent expects less benefit from acquiring information and proposing projects when a principal is likely to overrule her. Burkart et al. [1997] map this tradeoff to ownership. A larger block makes intervention worthwhile but can substitute the blockholder's judgment for the manager's and deter the noncontractible initiative only she can supply; the fixed-cost logic for why a sufficiently large owner bears the cost of control follows Shleifer and Vishny [1986]. We apply this control–initiative logic to attention to the future. In our extension, managerial equity matters because it changes the manager's return from a better project relative to her private benefit of control. We therefore derive and test an equity-conditioned concentration prediction for temporal reach.[4]

Exit is the leading alternative account. Prior theory asks how trading and voice interact in blockholder governance [Maug, 1998, Kahn and Winton, 1998, Faure-Grimaud and Gromb, 2004, Admati and Pfleiderer, 2009, Fos and Kahn, 2018, Edmans and Holderness, 2017], and empirical work shows that exit threats can transmit through managerial price sensitivity [Bharath et al., 2013, Chen et al., 2015, Dou et al., 2018].[5] That evidence makes liquidity-based exit terms and their interactions with managerial equity the relevant rivals to the intervention channel. Our framework differs by placing price discipline and intervention on the same forward-looking managerial action: breadth operates through price informativeness, while concentration can deter initiative, with inside equity governing the strength of that deterrence. The mechanism tests in Section 6 probe this conditional control–initiative account against the liquidity-based exit alternative.

[3] Russell reconstitution has also been used to study ownership and governance [Appel et al., 2016, Boone and White, 2015, Crane et al., 2016, Schmidt and Fahlenbrach, 2017].

[4] Related empirical work links longer executive pay and vesting horizons to long-term investment, firm value, and less earnings management [Gopalan et al., 2014, Edmans et al., 2017, Marinovic and Varas, 2019, Flammer and Bansal, 2017].

[5] Broader evidence documents the prevalence and heterogeneity of blockholders and their effects on monitoring, activism, firm value, and disclosure [Holderness, 2009, 2001, Cronqvist and Fahlenbrach, 2009, Clifford and Lindsey, 2016, Brav et al., 2008, Konijn et al., 2011, Ge et al., 2021].

3 Conceptual Framework and Predictions

This section develops a parsimonious framework that maps ownership structure into the manager's temporal reach. Two established governance forces shape that choice. First, informed trading makes the stock price more revealing, increasing the equity-linked return to actions whose payoffs lie in the future [Edmans and Manso, 2011]. Second, intervention by a large shareholder can improve project selection while weakening the manager's incentive to take initiative [Burkart et al., 1997]. We bring these forces together to distinguish three ownership margins—the level and breadth of institutional ownership, which govern price informativeness, and the blockholder's stake, which governs intervention. Related models study trading and intervention jointly from the blockholder's perspective [Maug, 1998, Kahn and Winton, 1998, Faure-Grimaud and Gromb, 2004]. The framework's purpose is to derive transparent predictions for a managerial choice those models do not study: how far ahead the firm looks.

3.1 Environment

The manager chooses a temporal reach $x \geq 0$: the distance into the future over which she searches for opportunities, evaluates their consequences, and plans for them. A larger x means looking further ahead, not exerting more effort over a fixed horizon. We normalize the private cost of reach to $x^2/2$; the results require only that this cost be increasing and convex. We call x the manager's *temporal reach*; it is the theoretical counterpart of the empirical *attention to the future* constructed in Section 4 (the months-ahead measure), and we reserve the term "temporal reach" for this model object.

A single firm has one unit of equity. The manager owns share $\mu \in (0, 1)$, $I \geq 1$ institutional owners hold stakes $s_1, \ldots, s_I$, and residual noninstitutional investors hold the float f. Total institutional ownership is β, and the blockholder is the largest institution, whose stake is m:

$$\beta = \sum_{i=1}^{I} s_i, \qquad m = \max_i s_i, \qquad \mu + \beta + f = 1. \tag{1}$$

The three ownership margins remain distinct. The level β and breadth I govern price informativeness; the largest block m governs control. We do not impose symmetry across institutional stakes, so m can vary independently of β and I within the feasible range $\beta/I \leq m \leq \beta$. This is the theoretical counterpart of entering level, breadth, and concentration jointly in the empirical analysis.

One unit of well-directed reach creates value $P > 0$. With probability $\lambda \in (0, 1)$, the manager's plan coincides with the value-maximizing plan. With probability $1 - \lambda$, the

manager and shareholders disagree, and value is created only if the manager is overruled. The manager also receives a private benefit $b \geq 0$ per unit of reach implemented under her own authority. Here, b is the private value of control and P is the shareholder value of the preferred project.

3.2 Price discipline and ownership breadth

Motivated by the trading channel of Edmans and Manso [2011], we summarize the pass-through of fundamental value into the price with a reduced-form informativeness index. Each of the I institutional owners is informed and has an independent signal with precision $q(\beta) > 0$. The restriction $q'(\beta) > 0$ is our extension: it allows a larger stock of institutional capital to support more information acquisition. The signal- aggregation interpretation in Appendix B.1 motivates the following index.

Lemma 1 (Price informativeness)**.** *Define the fraction of fundamental value impounded in the price as*

$$\Pi(\beta, I) = \frac{Iq(\beta)}{1 + Iq(\beta)} \in (0, 1) \tag{2}$$

of the value created by the manager's action. Hence $\partial\Pi/\partial\beta > 0$ *and* $\partial\Pi/\partial I > 0$*. When signal precision is fixed at* $q \equiv 1$*, the expression reduces to* $\Pi = I/(I + 1)$ *and isolates the breadth channel emphasized by Edmans and Manso [2011].*

The economic role of Π is to determine how faithfully value creation is reflected in the price on which the manager's equity wealth depends. Greater institutional ownership raises signal precision by assumption; greater breadth brings more independent information into the price. Both increase the manager's private return to looking further ahead. Because m is the stake of the largest institution, increasing it while holding total institutional ownership and breadth fixed does not raise Π in the benchmark: reallocating shares toward an already-informed owner adds no independent signal. Section 6 evaluates the principal alternative in which a block instead disciplines through informed exit.

3.3 Control and managerial initiative

When the manager and shareholders disagree, the largest owner can intervene and impose the value-maximizing plan. Intervention is a public good: all shareholders benefit, but the intervening owner alone bears the fixed organizational cost $K > 0$. An owner can remain inactive and earn zero or pay K and choose an intervention probability. Conditional on

entry and an interior probability, its problem is

$$\max_{0<E<1} \; sP(1-\lambda)xE - \frac{E^2}{2} - K. \tag{3}$$

The interior choice is $E = sP(1-\lambda)x$. Because its net value rises with s, and because a second intervention is redundant, only the largest owner may find it worthwhile to control.

Proposition 1 (The controlling blockholder)**.** *On the region in which the intervention cap is slack, at most one owner intervenes, and that owner is the largest, m. It enters if and only if*

$$mP(1-\lambda)x \geq \sqrt{2K}, \tag{4}$$

in which case $E = mP(1-\lambda)x$; otherwise $E = 0$.

The proposition supplies a microfoundation for concentration as the control margin in the spirit of Shleifer and Vishny [1986]. A sufficiently large owner internalizes enough of the gain to bear the fixed cost that smaller owners leave to others. Thus the model distinguishes an active block from a merely noncontrolling institutional owner without assuming that every firm has a controller. The entry comparison is against the zero-payoff outside option. The proof is in Appendix B.2; Appendix B.3 treats the corner at which the probability cap $E \leq 1$ binds.

The manager receives the private benefit of reach only when her plan survives, but she receives her equity share of firm value whether the initial plan survives or a better one is imposed. Given Π and E, she therefore solves

$$\max_{x \geq 0} \; bx[1-(1-\lambda)E] + \mu\Pi P x[\lambda + (1-\lambda)E] - \frac{x^2}{2}. \tag{5}$$

Her first-order condition is

$$x = (\mu\Pi P\lambda + b) + (1-\lambda)(\mu\Pi P - b)E, \qquad \frac{\partial x}{\partial E} = (1-\lambda)(\mu\Pi P - b). \tag{6}$$

Equation (6) isolates the manager's tradeoff. Intervention reduces the private benefit she obtains from retaining control, but raises the value of her equity when it replaces a lower-value plan. It therefore discourages temporal reach when $\mu\Pi P < b$ and encourages it when $\mu\Pi P > b$. This is the control–initiative trade-off of Burkart et al. [1997], recast here as a choice over temporal reach. Empirically, insider ownership provides a broad measure of the inside equity alignment represented by μ.

3.4 Equilibrium and comparative statics

Ownership is predetermined in the action subgame. Trading first fixes $\Pi(\beta, I)$; the manager then chooses x and an active block chooses E, each taking the other's action as given. We focus on an interior, active-block equilibrium.

Assumption 1 (Interior active-block equilibrium)**.** $\mu > 0$, $\Pi > 0$, the block clears the threshold in (4) strictly, the probability cap is slack in equilibrium ($mP(1-\lambda)x^* < 1$), and $1 - mP(1-\lambda)^2(\mu\Pi P - b) > 0$.

Proposition 2 (Equilibrium)**.** *Under Assumption 1, equilibrium reach and intervention are*

$$x^* = \frac{\mu\Pi P\lambda + b}{1 - mP(1-\lambda)^2(\mu\Pi P - b)}, \qquad E^* = mP(1-\lambda)x^*. \tag{7}$$

When $b = 0$, the equilibrium reduces to $x^*_{\text{price}} = \mu\Pi P\lambda/[1 - mP(1-\lambda)^2\mu\Pi P]$. When $\Pi = 0$ or $\mu = 0$, it reduces to $x^*_{\text{control}} = b/[1 + mP(1-\lambda)^2 b]$.

The two polar cases isolate the forces at work. In the first, only the manager's equity reward remains, so control and reach are complements. In the second, only the private benefit survives, so control and reach are substitutes. The interior equilibrium combines these two forces. The intervention cap prevents divergence when the equity channel is strong; complete derivations and the corner solution appear in Appendix B.3.

Proposition 3 (Ownership and attention to the future)**.** *In the interior equilibrium of Proposition 2:*

(i) Reach rises with the level and breadth of institutional ownership:

$$\frac{\partial x^*}{\partial \beta} > 0, \qquad \frac{\partial x^*}{\partial I} > 0.$$

(ii) The effect of blockholder concentration is conditionally signed:

$$\operatorname{sign}\left(\frac{\partial x^*}{\partial m}\right) = \operatorname{sign}(\mu\Pi P - b). \tag{8}$$

A larger block deters reach when private benefits dominate and encourages it when the manager's equity reward dominates.

(iii) A higher managerial equity stake raises the sign index $\mu\Pi P - b$. At the sign-change point $\mu\Pi P = b$, $\partial^2 x^*/(\partial m\, \partial\mu) > 0$. The cross-partial need not be positive away from that point, so the interaction prediction is local.

Appendix B.4 proves the proposition. The first result combines Lemma 1 with the manager's price-linked reward: ownership level and breadth both sharpen the price and thereby extend reach. The second and third results describe how managerial equity moderates the concentration effect. Near the sign-change point, a negative relation between block concentration and reach should attenuate as managerial equity rises. The data cannot locate firms relative to that unobserved point, so the interaction below is a reduced-form consistency test rather than identification of the structural cross-partial.

The framework maps directly into four empirical counterparts. The 13F institutional ownership share proxies for β, the number of institutional owners for I, the largest institutional owner's share for m, and insider ownership as a broad empirical counterpart for the inside equity alignment represented by μ. Price discipline predicts positive effects of institutional ownership and breadth. The intervention mechanism predicts a conditionally signed concentration effect and motivates a positive concentration–equity interaction locally near the sign-change point. These implications organize the empirical analysis. Productive engagement, firm value, and the value-maximizing ownership structure are developed in Appendix B.5; none is needed to derive the predictions for temporal reach. These are predictions for managerial attention, not welfare. A larger block can correct a poorly directed plan even as it deters initiative, so an ownership structure that maximizes reach need not maximize firm value.

4 Measuring Attention to the Future

The framework in Section 3 treats the distance over which a manager looks ahead as an economic choice. Taking its predictions to the data requires a measure of that distance that is direct, cardinal, and comparable across firms and over time. This section constructs such a measure and establishes the properties needed for the ownership tests that follow. We first explain its construction and interpretation, then describe its within-firm structure and evaluate its predictive validity and state dependence.

4.1 Construction and interpretation

We measure attention to the future—*future attention* for short— from earnings-call transcripts of U.S. public firms. For the main 2002–2020 sample, we identify each forward-looking statement, map each statement that references a datable horizon to that horizon, and take

the median over the $n_{i,t}$ dated statements in a firm-year:

$$FutureAttention_{i,t} = \text{median}\big(\, horizon_{i,t,1}, \ldots, horizon_{i,t,n_{i,t}} \,\big), \tag{9}$$

where the $n_{i,t}$ statements are those with an extractable horizon, each contributing one value with multiplicity; the median is undefined, and the firm-year excluded, when no statement carries a datable horizon. The measure is denominated in months: a value of five means that the typical forward-looking statement points about five months—roughly two quarters—ahead. The median captures where the firm's attention typically points while remaining insensitive to a handful of very-long-dated statements. [6] Appendix A documents construction and merging of the estimation panel.

The measure belongs to a broader text-as-data literature that recovers firm-level beliefs and attention from earnings calls [Gentzkow et al., 2019, Hassan et al., 2019, Sautner et al., 2023, Barrero, 2022]. Its closest antecedents infer corporate horizons from the relative incidence of short- and long-horizon language [Brochet et al., 2015], code the forward reach of calls in a small industry sample [DesJardine and Bansal, 2019], or hand-collect the stated duration of investment projects [Dessaint et al., 2023]. Our measure differs in yielding a cardinal firm-year quantity at scale: the number of months ahead to which managerial language typically points. Cardinal units matter because they turn the ownership coefficients below into economically interpretable changes in temporal reach.

Because the source is corporate communication, the principal measurement concern is that the statistic captures disclosure policy rather than attention. Forward-looking statements reflect litigation exposure and investor-relations effort [Li, 2010], and institutional ownership can increase both the quantity and precision of voluntary disclosure [Boone and White, 2015]. Construction around the median limits this quantity channel. Adding statements at the firm's prevailing reach leaves the measure unchanged; adding near-term guidance lowers it. Disclosure volume therefore cannot mechanically lengthen measured attention. The validation below supplies a distinct test of economic content: language that reaches further ahead should precede, rather than merely accompany, real activity whose returns arrive in the future.

Coverage	
Firm-years	50,375
Distinct firms	5,705
Distribution (months)	
Median	5.0
10th–90th percentile	2.6–7.5
Standard deviation	2.3
Structure	
Within-firm share of variance	55%
Persistence, AR(1)	0.44

Table 1: Coverage, distribution, and structure of attention to the future.

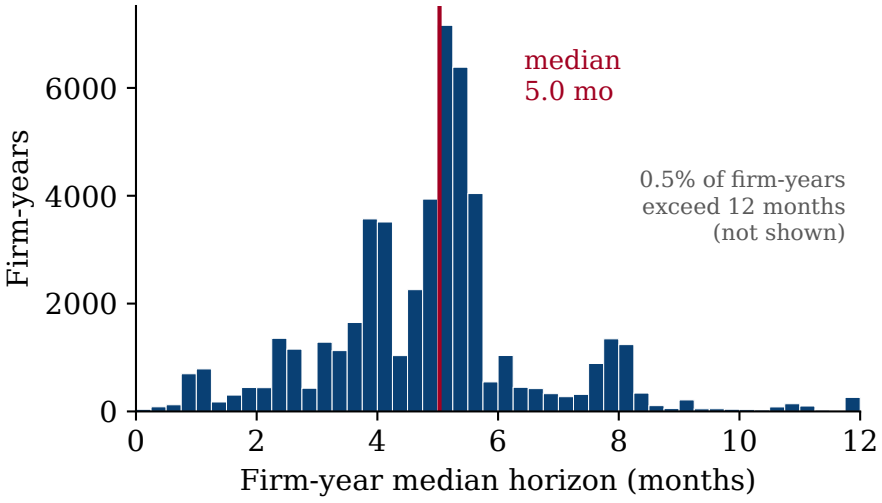


Figure 1: Distribution of firm-year attention to the future (months).

4.2 Distribution, within-firm variation, and persistence

Table 1 summarizes 50,375 firm-years representing 5,705 firms; Figure 1 plots their distribution. The median firm-year looks five months ahead, the 10th and 90th percentiles are 2.6 and 7.5 months, and the standard deviation is 2.3 months. The five-month center is natural for a quarterly conversation: the typical dated statement concerns the next quarter or two. The economically useful content lies in the dispersion around that center—whether a given firm-year looks three months ahead or eight.

The variation is not simply a permanent firm characteristic. Fifty-five percent of the variance is within firm, leaving substantial movement for the firm fixed-effects designs below to exploit. At the same time, an AR(1) coefficient of 0.44 shows that the measure contains persistent signal rather than only transitory variation. This combination—meaningful persistence and considerable within-firm movement—is what one would expect of an organizational choice that reflects both durable firm attributes and changes in incentives or circumstances.

Because the ownership designs of Section 5 identify their effects from precisely this within-firm variation, we standardize the outcome there: rather than reporting coefficients in months, we express attention in units of its within-firm standard deviation (one such standard deviation is about 1.35 months in the ownership sample; see Table 4), and quote the resulting effects in the text as a percentage of a within-firm standard deviation.

4.3 Predictive validity

If the measure captures how far ahead a firm is actually looking, firms that look farther ahead today should subsequently undertake more activity whose payoffs require advance preparation. Table 2 shows that they do. We regress five one-year-ahead (t+1) outcomes on the standardized measure, over fiscal years 2002–2024,[7] absorbing sector×year fixed effects and controlling for firm size (log lagged assets, and additionally Tobin's Q in the investment column); outcomes and the measure are winsorized at the 1st and 99th percentiles and standard errors are clustered by firm. The outcomes trace the progression from expenditure to realized innovation: capital expenditure and R&D from Compustat (each scaled by lagged assets), the growth of firm patent output, and—because looking ahead should raise the quality of what a firm eventually produces, not only the quantity—the market value and

[6] We reserve *horizon* for the dated reach of an individual statement. The firm-year median is the central tendency of that statement-level distribution—where the firm's language typically points—and we call it attention to the future throughout.

[7] Table 1 and the ownership analysis end in fiscal 2020 to match the ownership data; the validation needs no ownership variables and uses the full transcript span.

citation quality of its subsequent patents, measured with the stock-market-based patent valuations of Kogan et al. [2017].

Dependent variable	asinh(CAPX / lag assets)	asinh(R&D / lag assets)	IHS patent growth	asinh(KPSS issue value, mean)	KPSS filing high-cite share
	(1)	(2)	(3)	(4)	(5)
Attention to the future	0.256***	0.840***	0.0071**	0.0575***	0.0040***
	(0.035)	(0.123)	(0.0030)	(0.0117)	(0.0014)
Log lag assets control	Yes	Yes	Yes	Yes	Yes
Tobin's Q control	Yes	No	No	No	No
SIC2 × year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	44,749	27,708	26,441	16,137	44,794
Firms	5,245	3,388	4,215	2,374	5,372

Notes: Regressions of one-year-ahead (t+1) outcomes on standardized attention to the future (F_{it}, `horizon_p50y`), with SIC2×year fixed effects. Outcomes and F_{it} are winsorized at the 1st and 99th percentiles inside each estimation sample; all columns control for log lag assets, and the investment column additionally for Tobin's Q. CAPX and R&D coefficients and standard errors are multiplied by 100 (percentage points of the outcome ratio). Standard errors clustered by firm in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 2: Attention to the future predicts real forward-looking activity at t+1.

All five coefficients are positive and statistically significant. A one-standard-deviation increase in attention to the future is followed a year later by capital expenditure higher by about 0.25 percentage points of assets and R&D higher by about 0.8 percentage points—roughly 6 and 9 percent of their respective sample means—together with faster patent growth and higher patent value and citation quality. For scale, a one-standard-deviation increase in firm-level political risk—a leading text measure built from the same earnings-call corpus—moves the investment rate by 0.16 percentage points, about 1.4% of its sample mean, in a comparable sector×time specification [Hassan et al., 2019]; the attention gradient, at roughly 6% of mean capital expenditure, is of the same order or larger. The measure thus forecasts real activity that unfolds over the following year, as a measure of attention to the future should.

Two features guard against a mechanical reading. First, the outcomes are dated t+1: they describe what a firm does *after* it speaks, not a contemporaneous accounting relationship. Second, the associations survive the more demanding within-firm comparison. Appendix Table 9 re-estimates each column with firm fixed effects added, so that identification comes only from a firm's own variation in attention over time: the capital-expenditure, R&D, patent-value, and patent-quality results remain positive and significant—only patent-count growth attenuates to insignificance—and the investment result is robust to controlling for the firm's cost of capital and perceived discount rate [Gormsen and Huber, 2025]. Those estimates say something specific: higher measured attention within a firm is followed by higher investment one year later. The outcome variables and their sources are described in Appendix A.

4.4 Financial distress and state dependence

A useful measure should also move with economic circumstances. Financial pressure raises the value of resolving immediate problems and tightens the resources available for preparing for distant ones; financial slack relaxes that constraint. Table 3 shows that the measure behaves this way within firms. Each row is a separate regression of the year-on-year change in a firm's future attention on one lagged stressor, with firm and sector×year fixed effects and standard errors clustered by firm, so each estimate describes how a firm's own attention moves when its situation changes relative to its industry peers. Because the dependent variable is itself a year-on-year change, the firm effects absorb firm-specific *trends* in attention rather than firm levels, so each estimate is the response of the change to a lagged stressor, net of the firm's own trend and of industry-year shocks. The stressors pair accounting and market distress from Compustat and CRSP—an operating loss, high leverage, a low cash ratio, a stock-price crash, and a market-based distance-to-default—with an earnings miss relative to the I/B/E/S analyst consensus and an overall-risk index built from the firm's own earnings-call text following Hassan et al. [2019] and provided by NL Analytics. Because this last index shares its source with the attention measure, we lag it—as we do every stressor—to avoid mechanical overlap with the contemporaneous measure.

	Dependent variable: Δ attention to the future (months)	
Stressor_{t-1}	Estimate (SE)	Obs.
Overall risk	-0.045^{***} (0.012)	41,846
Earnings miss	-0.053^{**} (0.018)	34,492
Stock-price crash ($<-30\%$)	-0.116^{***} (0.026)	40,667
Operating loss (EBIT<0)	-0.101^{***} (0.032)	41,846
Net leverage	-0.079^{***} (0.017)	41,651
Cash ratio	$+0.045^{***}$ (0.016)	35,328
Distance-to-default	$+0.058^{***}$ (0.012)	34,450

Notes: Each row is a separate regression of the first difference of the firm-year median attention to the future (months) on one lagged stressor. Earnings miss, stock-price crash, and operating loss are indicators (per event); the remaining regressors are standardized (per SD). All include firm and sector×year fixed effects; SE (in parentheses) clustered by firm; 2002–2020. $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Within-firm response of attention to the future to prior-year financial stress.

The estimates have a simple summary: *firms in trouble pay less attention to the future.* In the year after a firm faces elevated risk, an earnings miss, a stock-price crash, an operating loss, or high leverage, its future attention contracts; after it accumulates cash or moves away from distress—a higher distance-to-default—it extends. This uniform pattern both

validates the measure and documents systematic state dependence with respect to financial distress. The individual effects are modest—a stock-price crash contracts attention by about 0.12 months, the largest of the per-event responses—but they are precisely estimated and uniformly signed. Appendix A gives the construction and source of each variable.

Taken together, the distribution, predictive-validity, and state-dependence evidence establish that the measure captures economically meaningful variation in firms' temporal reach rather than merely the volume of corporate communication.

5 Ownership Structure and Attention to the Future

Having established that attention to the future is measurable, persistent, and economically meaningful, we now test the framework's predictions for ownership. We first estimate the within-firm relations between attention and three distinct ownership margins: the level of institutional ownership, its breadth across investors, and its concentration in the largest block. We then use Russell index assignment as a supporting direction-of-causality test: it uses index-driven variation in passive ownership and asks whether attention subsequently moves in the predicted direction.

5.1 Data and specification

Ownership data. Our measures come from the Ownership Project [Amel-Zadeh et al., 2022], which reconstructs owner-level holdings in U.S. public firms from Forms 13F, Schedules 13D and 13G, and insider Forms 3, 4, and 5. This integrated coverage is the source's key advantage for our analysis: institutional investors, blockholders, and insiders can be measured consistently within the same firm and period. It reports both shares and votes at the owner–firm–quarter level. Its raw file contains roughly 58 million positions from 2000Q4 through 2020Q4. For each firm-quarter, and then averaged over the calendar year, we construct the 13F institutional share, the number of distinct institutional owners (breadth), the largest institutional owner's share (concentration), and indicators for blockholder stakes above 5%, 10%, and 20%. We remove positions flagged as outliers, retain records supported by a 13F, 13D, or 13G filing, and consolidate multiple positions filed by the same owner in a firm-quarter. The baseline ownership-level measure uses 13F positions, while the block and concentration measures draw on the broader 13F, 13D, and 13G universe. Holdings from Forms 3, 4, and 5 provide the broad measure of inside equity alignment used in Section 6. Corporate officers are the managers most directly involved in choosing and communicating firm policy; directors and 10-percent owners broaden the measure to other statutory insiders.

Appendix C provides the variable definitions and the broader set of ownership measures used in the robustness analysis.

Sample. We merge the firm-year ownership measures with the attention-to-the-future measure of Section 4 and with firm fundamentals from Compustat. Compustat supplies the size control (lagged book assets), the SIC industry classification used to form the sector×year effects, and the balance-sheet variables—sales, leverage, cash flow, cash holdings, and Tobin's Q—used in the appendix robustness checks. The attention measure is matched to Compustat by firm identifier (GVKEY), and the ownership data are matched to the panel by firm (PERMCO) and fiscal year. The two identifiers are bridged with the CRSP/Compustat Merged (CCM) link, applied at the fiscal-year period-end date; because ownership is a calendar-year average, it is assigned to the fiscal year it most nearly spans, an approximation for the minority of firms whose fiscal year does not end in December. The estimation sample is the intersection of firms with a transcript-based attention value, an ownership record, and a non-missing size control: 35,416 firm-years covering 3,593 distinct firms and 1,207 sector-years over fiscal years 2002–2020. Table 4 reports summary statistics for the variables that enter the analysis.

	N	Mean	SD	Within SD	p10	Median	p90
Dependent variable							
Attention to the future (months)	35,416	4.93	1.83	1.35	2.73	5.05	7.37
Ownership structure							
13F institutional ownership share	35,416	0.842	0.272	0.138	0.456	0.882	1.154
Breadth (no. of institutional owners)	35,416	306.5	326.1	114.1	59	202	668
Largest institutional-owner share	35,416	0.136	0.095	0.058	0.067	0.108	0.233
Control							
Log lagged book assets	35,416	7.11	2.06	0.52	4.56	6.98	9.85

Notes: The estimation sample is the intersection of firms with a transcript-based attention-to-the-future value, an ownership record, and a non-missing size control: 35,416 firm-years, 3,593 firms, and 1,207 sector-years over fiscal years 2002–2020—the sample of the three-fact regression (Table 5). Attention to the future is the firm-year median forward-looking horizon in months. The 13F share is summed reported 13F holdings relative to shares outstanding and is not top-coded; it can exceed one when filing dates and the shares-outstanding denominator do not align. The largest institutional-owner share is also expressed relative to shares outstanding. Breadth is the number of distinct institutional owners and enters the regressions in logs. The Within SD column reports the within-firm standard deviation: the standard deviation of each variable after subtracting its firm-specific mean, the dispersion the firm fixed effects leave available for identification. The ownership regressors are standardized within the sample in the regressions; the statistics reported here are in these natural units. The regression tables that follow report coefficients in units of the dependent variable's within-firm standard deviation; this column gives the conversion, as one within-firm SD of attention is 1.35 months.

Table 4: Summary statistics for the ownership estimation sample.

Specification. We relate attention to the future to ownership structure through the two-way fixed-effects regression

$$FutureAttention_{i,t} = \alpha_i + \gamma_{j,t} + \beta_1 \,\text{13F-Share}_{i,t} + \beta_2 \,\text{Breadth}_{i,t} + \beta_3 \,\text{Blockholder}_{i,t} + X'_{i,t-1}\theta + \varepsilon_{i,t}, \tag{10}$$

where $\text{13F-Share}_{i,t}$ is the 13F institutional ownership share, $\text{Breadth}_{i,t}$ is the log number of distinct institutional owners, and $\text{Blockholder}_{i,t}$ is the largest institutional owner's share; α_i is a firm fixed effect, $\gamma_{j,t}$ a sector×year fixed effect, and $X_{i,t-1}$ contains log lagged assets. The ownership regressors are standardized within the estimation sample, and each coefficient is expressed in units of the dependent variable's within-firm standard deviation, where one within-firm SD of attention is 1.35 months (Table 4). In the text we quote these effects as a percentage of a within-firm standard deviation, so that a table coefficient of, say, 0.07 is a 7% change; each reports the response to a one-standard-deviation change in the corresponding ownership variable. Standard errors are clustered by firm.

Identifying variation. Firm effects remove permanent differences in business model, governance, and culture; sector×year effects remove shocks shared by an industry in a given year. The baseline coefficients therefore describe how changes in a firm's ownership structure co-move with changes in its attention, relative to its industry peers. Section 5.3 then complements these estimates with index-induced passive ownership, asking whether an ownership change not selected on the firm's prior language precedes a change in attention. This is a useful division of labor in ownership research, where credible instruments for block ownership are rare and clean designs typically isolate a single ownership margin [Edmans and Holderness, 2017]. Clustering by firm allows the errors to be arbitrarily correlated over time within a firm.

5.2 Baseline ownership evidence

Table 5 reports the empirical counterparts of the three ownership predictions. Column (1) enters institutional ownership alone; columns (2) and (3) add breadth and then the blockholder share, so that the full specification isolates each margin holding the other two fixed.

The first fact is that *attention to the future rises with institutional ownership.* A one-standard-deviation increase in the 13F share raises attention by about 7% of a within-firm standard deviation (roughly a tenth of a month) in column (1). The estimate remains positive and statistically significant when breadth and blockholder concentration are held fixed, at about 5.5% of a within-firm standard deviation in column (3). This magnitude is of the same order as the response of attention to a one-standard-deviation change in the firm's overall

	Dependent variable: attention to the future (within SD)		
	(1) Level	(2) +Breadth	(3) +Block
Institutional ownership (13F)	0.071***	0.047***	0.055***
	(0.015)	(0.016)	(0.016)
Breadth (log # owners)		0.145***	0.111***
		(0.036)	(0.038)
Blockholder (largest institutional-owner share)			−0.037***
			(0.012)
Firm FE	Yes	Yes	Yes
Sector × year FE	Yes	Yes	Yes
Observations	35,416	35,416	35,416
Firms	3,593	3,593	3,593
Within R^2	0.003	0.004	0.004

Notes: Institutional ownership is the 13F share; breadth is the log number of institutional owners; blockholder is the largest institutional owner's share. Firm and sector×year fixed effects; ownership standardized within sample. Coefficients are expressed in within-firm standard deviations ("within SD") of attention to the future per one-SD increase in the regressor (one within-firm SD ≈ 1.35 months; see Table 4); SE clustered by firm. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Ownership structure and attention to the future.

risk, leverage, or distance to default (Table 3). The result complements evidence that investor horizons matter for corporate language and policy [Bushee, 1998, Brochet et al., 2012]: here the relation survives within firm, net of permanent firm attributes and sector–year shocks, while holding breadth and concentration fixed.

The second fact is that attention rises with ownership breadth. Conditional on the institutional share, a firm held by more distinct institutions looks 11–15% of a within-firm standard deviation farther ahead per standard deviation of breadth (columns 2–3), the largest of the three margins. Breadth absorbs part of the level effect, consistent with the number of informed investors mattering beyond their aggregate stake.

The third fact runs in the opposite direction. A one-standard-deviation increase in the largest institutional owner's share is associated with a contraction of about 3.7% of a within-firm standard deviation in attention (column 3), comparable to the contraction following an earnings miss. This result complements evidence that blockholder identity and activism shape corporate policy and disclosure [Brav et al., 2008, Clifford and Lindsey, 2016, Guo et al., 2024, Ge et al., 2021].

More broadly, prior work distinguishes transient or high-churn institutions from long-horizon owners [Bushee, 2001, Gaspar et al., 2005, Yan and Zhang, 2009] and links patient ownership to investment, innovation, and monitoring [Derrien et al., 2013, Bena et al., 2017, Aghion et al., 2013, Harford et al., 2018]. Our three regressors instead distinguish the level, breadth, and concentration of institutional ownership within the same firm.

The estimates are precise despite a noisy dependent variable. Attention is a firm-year median over a finite set of forward-looking statements. To the extent that the resulting sampling error is mean-independent of ownership, it enters the outcome rather than the regressors: it increases residual variance without attenuating the coefficients and therefore

works against statistical precision.

These patterns are robust. Tables 12 and 13 trace the institutional-ownership association across specifications: it holds with and without firm fixed effects, when the wedge between voting and cash-flow rights is added, when current and lagged ownership enter jointly—where the contemporaneous share carries the effect and the lag is flat—and when the size control is replaced by a fuller set of lagged fundamentals (sales, leverage, cash flow, and cash holdings). Decomposing ownership by filing source locates the positive association in 13F investment-manager ownership, while 13D active-blockholder ownership enters negatively—an early trace of the concentration fact. Table 14 shows that the concentration fact does not depend on how it is measured: holding total institutional ownership fixed, attention falls with the largest institutional owner's share, the gap to the second-largest owner, an owner Herfindahl index, blockholder presence above 20%, the share held by owners above 10% or 20%, and the combined 13D/13G blockholder share—every measure negative and significant at the 1% level.

5.3 Russell index assignment: a direction-of-causality test

We complement the within-firm estimates with variation induced by the annual reconstitution of the Russell 1000 and Russell 2000 indexes. The design was introduced as an instrument for institutional ownership by Appel et al. [2016] and has been used widely since [Boone and White, 2015, Crane et al., 2016, Schmidt and Fahlenbrach, 2017]. Each May, Russell ranks firms by market capitalization and assigns the largest 1000 to the Russell 1000 and the next 2000 to the Russell 2000. Because the indexes are capitalization-weighted, a firm near the cutoff receives a far larger portfolio weight at the *top* of the Russell 2000 than at the *bottom* of the Russell 1000, and index-tracking funds must hold it in proportion to that weight. Firms near the threshold are similar in size but differ sharply in the passive ownership they attract. Conditional on the running variable—market- capitalization rank—and firm size, the assignment therefore generates quasi-random variation in index-fund demand.

We use assignment to the Russell 2000 as an instrument for a firm's passive ownership.[8] Table 6 reports the estimates. The sample covers the post-2007 reconstitutions;[9] we compare firms within ±500 rank positions of the cutoff and include firm and sector×year fixed

[8]The ownership interpretation assumes that, conditional on the assignment controls, index assignment affects attention through the induced change in passive holdings.

[9]Beginning in 2007, FTSE Russell adopted a *banding* rule to curb index turnover: a firm near the cutoff retains its prior-year index unless its market capitalization moves beyond a band of ±2.5% of the cumulative market capitalization of the Russell 3000E around the Russell 1000/2000 breakpoint. Index membership near the cutoff therefore depends on a firm's prior-year assignment, not only its current rank, and we construct the instrument to reflect these rules [Appel et al., 2019].

effects. The first stage is strong: crossing into the Russell 2000 raises passive ownership by about 1.1 percentage points, with a first-stage F-statistic of 103, squarely in the range reported by the index-reconstitution literature. It is not an artifact of a few large index funds: excluding the largest trackers leaves the passive-ownership first stage intact (Appendix Table 15). The second stage estimates the response of attention to that index-induced passive ownership. The timing is consistent with passive ownership affecting attention rather than the reverse. The estimates in the pre-period $(t-1)$ and year of reassignment are statistically indistinguishable from zero; the effect is positive and statistically significant at $t+1$, when a one-percentage-point increase in passive ownership raises attention by 0.17 months. Because the instrumented regressor is an unstandardized ownership share, this section reports magnitudes in natural units—months of attention per percentage point of passive ownership—rather than the within-firm standard-deviation units of the preceding tables. The null estimate on impact is partly mechanical: the outcome is a fiscal-year median of the firm's calls and reconstitution occurs at the end of June, so roughly half of the year-t calls predate the ownership change—t+1 is the first fully exposed year. The $t+1$ estimate is nearly unchanged when firm-size controls are added and remains positive and significant, though larger, when the window is narrowed to ±250 ranks.

	First stage	Second stage (2SLS)				
	passive own. (pp)	attention to the future (months)				
		event time			robustness t+1	
		$t-1$ *(pre)*	t	t+1	+size	±250
Russell 2000 assignment	+1.060***					
	(0.104)					
Passive ownership		−0.002	+0.040	+0.171***	+0.167***	+0.265**
		(0.056)	(0.057)	(0.057)	(0.053)	(0.110)
First-stage F	103					
Observations	10,090	10,069	10,564	10,090	10,040	4,793

Notes: Col. 1 is the first stage (Russell 2000 assignment → S12 passive ownership, pp, t+1 cell); cols. 2–6 are 2SLS (months of attention to the future per +1pp). Firm + SIC2×year FE, quad. ln(cap), ±500 ranks, 2007–2020, firm clusters. Robustness adds firm-size controls (+size) or narrows to ±250 ranks. The instrument moves only the index/passive margin, not total institutional ownership. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Russell index-assignment IV: passive ownership and attention to the future.

Thus, when passive ownership moves for reasons external to the firm's prior language, attention moves in the direction predicted by the baseline ownership results. The full instrumental-variable and reduced-form specifications appear in Appendix Table 15, together with first-stage diagnostics, alternative windows, and additional fixed-effect structures. Appendix Table 16 adds standard reconstitution diagnostics—running-variable density and the sensitivity of the estimate across windows, fixed effects, and clustering.

6 Testing the Concentration–Equity Prediction

The preceding evidence establishes how attention to the future varies with the level, breadth, and concentration of ownership. We now test whether insider ownership moderates the effect of blockholder concentration in the direction predicted by the model. We then ask whether a liquidity-based exit mechanism can account for the same pattern.

6.1 The blockholder–inside-equity interaction

The empirical counterpart of the model's largest block m is the largest institutional owner's share. We use ownership reported on Forms 3–5 as a broad empirical proxy for inside equity alignment. Crucially, these filings cover corporate officers—the managers who choose and communicate firm policy—alongside directors and 10-percent beneficial owners, yielding a consistent firm-level measure of equity held by statutory insiders. This broad measure lets us test the alignment mechanism. The model predicts that this alignment attenuates the initiative cost imposed by intervention, yielding a positive $m \times \mu$ interaction in the conflict region. Table 7 builds up the specification: column (1) includes the raw interaction, and columns (2)–(4) add 13F institutional ownership, breadth, and firm size in turn.

	Dependent variable: attention to the future (within SD)			
	(1) Raw	(2) $+\beta$	(3) $+N$	(4) +Size
Blockholder m	-0.057^{***}	-0.054^{***}	-0.046^{***}	-0.048^{***}
	(0.013)	(0.013)	(0.014)	(0.014)
Insider ownership μ	-0.025	-0.032^{*}	-0.022	-0.023
	(0.017)	(0.017)	(0.017)	(0.017)
$m \times \mu$	0.017^{***}	0.015^{**}	0.014^{**}	0.014^{**}
	(0.006)	(0.006)	(0.006)	(0.006)
Institutional ownership β	–	Yes	Yes	Yes
Breadth N	–	–	Yes	Yes
Size (log market cap)	–	–	–	Yes
Firm FE	Yes	Yes	Yes	Yes
Sector × year FE	Yes	Yes	Yes	Yes
Observations	35,416	35,416	35,416	35,399
Firms	3,593	3,593	3,593	3,592

Notes: Build-up of the block–equity interaction. Column (1) is the raw interaction; columns (2)–(4) add 13F institutional ownership β, breadth N (log number of owners), and firm size (log market cap) one at a time. m largest single institutional owner's share; μ insider ownership (Forms 3/4/5: officers, directors, 10%+ owners); $m \times \mu$ their interaction (product of standardized components). Firm and sector×year fixed effects and a lagged-book-assets control throughout; ownership standardized within sample. Coefficients are expressed in within-firm standard deviations ("within SD") of attention to the future (one within-firm SD $\approx$ 1.35 months; see Table 4): main-effect coefficients per one-SD increase in the regressor, interaction coefficients per one-unit increase in the product of standardized components. SE clustered by firm. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Blockholder concentration and insider ownership.

The interaction is positive and significant throughout, and remains stable as the controls enter: $m \times \mu$ moves only from 1.7% to 1.4% of a within-firm standard deviation of attention per unit of the product of standardized m and μ, while the block's average effect remains

negative (−4.6% to −5.7% per one-standard-deviation increase in the block). At the fully controlled estimates, a one-standard-deviation increase in the insider-ownership measure reduces the block's deterrent effect by about 30 percent. A two-standard-deviation increase nearly halves it, from −6.1% to −3.4% of a within-firm standard deviation per standard deviation of the block. The effect attenuates but does not reverse: it remains negative and significant throughout the observed support of insider ownership. The interaction is also robust to absorbing nonlinearities in block size and to residualizing insider ownership with respect to block size (Appendix B.6).

This attenuation is the predicted moderation of the concentration effect. The test uses within-firm changes in equilibrium ownership, and the principal exit-based alternative does not absorb it, as we show next.

6.2 A liquidity-based exit alternative

The positive block–equity interaction does not by itself identify control. In exit theories, governance also operates through the manager's price-linked wealth, so a blockholder's threat of informed selling can become more powerful as managerial equity rises. Bharath et al. [2013] find that the relation between blockholdings and firm value responds to exogenous liquidity shocks, especially when managerial wealth is sensitive to the stock price; and Dou et al. [2018] report the corresponding result for earnings management. The same logic could generate the interaction in Table 7 without a control channel.

Exit and control differ in their dependence on market liquidity. Exit requires a credible threat of informed selling and is therefore stronger in liquid stocks [Admati and Pfleiderer, 2009, Edmans, 2009]. Indeed, Bharath et al. [2013] identify this channel from the interaction of blockholdings with liquidity and find it concentrated among price-sensitive managers. Direct intervention requires no trade. If exit explains our block–equity coefficient, that coefficient should be absorbed by liquidity interacted with insider ownership, or by liquidity interacted jointly with the block and insider ownership. Under the control interpretation, neither pattern is required.

Table 8 reports the horse race. Column (1) repeats the specification of Table 7 in the subsample with CRSP coverage. The remaining columns add two liquidity measures: L, the negative logarithm of the Amihud [2002] illiquidity ratio, and T, log share turnover. Columns (2)–(4) interact these measures with insider ownership; columns (5)–(6) add their interactions with the block and with the block–equity term itself.

Three results emerge. First, the exit-threat terms leave the block–equity interaction essentially unchanged. Across the six columns, $m \times \mu$ ranges from 1.2% to 1.7% of a within-

	Dependent variable: attention to the future (within SD)					
	(1) Baseline	(2) +Liquidity	(3) +Turnover	(4) Both	(5) Liq. full	(6) Turn. full
Blockholder m	-0.045^{***}	-0.044^{***}	-0.043^{***}	-0.043^{***}	-0.042^{***}	-0.041^{***}
	(0.014)	(0.014)	(0.014)	(0.014)	(0.016)	(0.015)
Insider ownership μ	-0.026	-0.029	-0.029^{*}	-0.026	-0.030	-0.033^{*}
	(0.017)	(0.019)	(0.017)	(0.019)	(0.019)	(0.018)
$m \times \mu$	0.015^{**}	0.015^{**}	0.012^{*}	0.013^{*}	0.014^{*}	0.017^{**}
	(0.006)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Liquidity L		0.010		0.012	0.010	
		(0.037)		(0.041)	(0.037)	
$L \times \mu$		-0.007		0.007	-0.007	
		(0.011)		(0.013)	(0.015)	
$L \times m$					0.007	
					(0.013)	
$L \times m \times \mu$					-0.002	
					(0.006)	
Turnover T			0.003	0.001		0.001
			(0.014)	(0.015)		(0.014)
$T \times \mu$			-0.017^{**}	-0.020^{**}		-0.028^{***}
			(0.008)	(0.009)		(0.010)
$T \times m$						0.003
						(0.009)
$T \times m \times \mu$						0.007
						(0.004)
β, breadth N, size, lagged assets	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector × year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	35,030	35,030	35,030	35,030	35,030	35,030
Firms	3,555	3,555	3,555	3,555	3,555	3,555

Notes: Estimates of the block–equity interaction alongside the exit-threat channel of Bharath, Jayaraman, and Nagar (2013), which operates through stock liquidity interacted with the manager's price-linked wealth. L is liquidity, the negative log of the Amihud (2002) illiquidity ratio (monthly |return| per dollar volume from CRSP, averaged over the year); T is log share turnover (monthly volume over shares outstanding, averaged over the year). Column (1) repeats the baseline interaction specification on the CRSP-merged sample; (2)–(4) add each liquidity measure and its interaction with the managerial-equity proxy μ; (5)–(6) add the interactions with the block m and with $m \times \mu$. All columns include 13F institutional ownership β, breadth N, size (log market cap), a lagged-book-assets control, and firm and sector×year fixed effects; ownership and liquidity standardized within sample. Coefficients are expressed in within-firm standard deviations ("within SD") of attention to the future (one within-firm SD $\approx$ 1.35 months; see Table 4): main-effect coefficients per one-SD increase in the regressor, interaction coefficients per one-unit increase in the relevant product of standardized components. SE clustered by firm. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: The block–equity interaction and the liquidity-based exit threat.

firm standard deviation per unit of the product of standardized m and μ, against a baseline of 1.5%, and remains statistically significant throughout. Second, the exit terms themselves are either indistinguishable from zero or have the opposite sign. Liquidity interacted with insider ownership is -0.7% (s.e. 1.1%), while turnover interacted with insider ownership ranges from -1.7% to -2.8% and is statistically significant. Attention to the future is therefore lower, not higher, where insiders have greater equity exposure and the stock is heavily traded. This pattern is inconsistent with simple exit-based discipline and accords with evidence that high-turnover ownership accompanies short-horizon corporate language [Brochet et al., 2012]. Third, the triple interactions $L \times m \times \mu$ and $T \times m \times \mu$ are small and insignificant. The moderating role of insider ownership is no stronger in liquid than in illiquid stocks, although an exit account predicts precisely such a gradient.

A complementary restriction follows directly from the price channel. Any block effect that reaches the manager only through the price enters her marginal return through $\mu\Pi$. Its level effect and its interaction with inside equity must therefore share the sign of $\partial\Pi/\partial m$. A block that improves price discovery raises reach, especially for an equity-paid manager; a block that crowds out trading competition lowers both. The estimates instead combine a negative level effect with a positive equity interaction. A mechanism operating only through the price does not naturally deliver that pairing. It is consistent with a control cost outside the price—the deterrence of initiative—that greater equity exposure partially offsets. These tests sharpen the control interpretation of the block–insider-equity interaction.

7 Conclusion

This paper introduces a cardinal measure of firms' revealed attention to the future. The measure reads the temporal reach of forward-looking language in earnings calls and expresses it in months. It captures a margin that investment, R&D, and patent outcomes cannot isolate: how far ahead management looks before those actions are chosen. Its behavior supports that interpretation. The measure is persistent but varies substantially within firms, predicts subsequent investment and innovation, and contracts when firms encounter financial distress.

We use the measure to study the relation between ownership and managerial temporal reach. Attention to the future increases with the level and breadth of institutional ownership and decreases with blockholder concentration. Russell index assignment supplies a complementary direction-of-causality test: when passive ownership rises for reasons external to the firm's prior language, attention subsequently extends. The concentration result, in turn, is consistent with the model's intervention mechanism: the blockholder's negative association

weakens as insider ownership rises. Liquidity and turnover interactions do not absorb that pattern, making a simple exit-based explanation difficult to reconcile with the evidence.

The Ownership Project's integrated coverage of institutions, blockholders, and insiders is central to this evidence: it lets us study institutional ownership structure and inside equity alignment jointly. Taken together, the results show that ownership is related not only to which projects a firm ultimately undertakes, but also to the temporal reach with which management searches, evaluates, and plans. Corporate governance therefore shapes an input into long-run action that precedes the actions conventionally observed.

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A Data and Measure Construction

This appendix documents the estimation panel behind Section 4: how the attention-to-the-future measure is merged to firm fundamentals, how the validation outcomes (Table 2) and financial-stress variables (Table 3) are constructed, and where each comes from. The measure itself—the transcript corpus and the horizon-extraction algorithm—is described in Section 4.1; the ownership data are described in Section 5.1 and Appendix C.

Sample construction and merge. We take Compustat annual as the master panel and merge the firm-year attention measure to it on GVKEY, matching a firm's calendar-year calls to the fiscal year of the same label; the merge is one-to-one and preserves the Compustat row count, and a firm-year enters the estimation samples only when it has a non-missing attention value and the size control. Over 2002–2020 the measure is available for roughly 50,000 firm-years and 5,700 firms. Firm identifiers are linked across sources with the CRSP/Compustat Merged (CCM) fiscal-year bridge, which attaches a PERMCO to each GVKEY–fiscal-year; this link is what allows the PERMCO-keyed patent-value and ownership files to be joined to the Compustat master. Because same-year Compustat items are not predetermined with respect to same-year attention, the validation and ownership specifications lag the controls and date outcomes at $t+1$.

Validation outcomes. The outcomes in Table 2 come from two kinds of source. Capital expenditure and R&D are from Compustat, each scaled by lagged book assets, as are the size control (log lagged assets) and Tobin's Q. The innovation outcomes combine two patent datasets: patent counts, and their growth, are drawn from a firm-year patent panel keyed by GVKEY, while the value and citation quality of patents are taken from the stock-market-based patent valuations of Kogan et al. [2017], linked to firms by PERMCO. From the latter we use the mean real value of the patents a firm is issued in a year and the share of its filings that are highly cited. All outcomes and the measure are winsorized at the 1st and 99th

percentiles within each estimation sample; the capital-expenditure and R&D coefficients are reported in percentage points of the outcome ratio.

Table 9: Validation of attention to the future with firm fixed effects.

Dependent variable	asinh(CAPX / lag assets)	asinh(R&D / lag assets)	IHS patent growth	asinh(KPSS issue value, mean)	KPSS filing high-cite share
	(1)	(2)	(3)	(4)	(5)
Attention to the future	0.137***	0.166**	0.0069	0.0170**	0.0010
	(0.024)	(0.066)	(0.0047)	(0.0085)	(0.0009)
Log lag assets control	Yes	Yes	Yes	Yes	Yes
Tobin's Q control	Yes	No	No	No	No
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
SIC2 × year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	44,197	27,332	25,915	15,663	44,210
Firms	4,694	3,016	3,689	1,926	4,788
Within R^2	0.0522	0.0525	0.0004	0.0324	0.0004
Mean dependent variable	0.0420	0.0929	-0.0078	14.62	0.0573

Notes: One-year-ahead (t+1) outcomes on standardized attention to the future (F_{it}, `horizon_p50y`). All columns include firm and SIC2×year fixed effects. Outcomes and F_{it} are winsorized at the 1st and 99th percentiles; all columns control for log lag assets, and the investment column additionally for Tobin's Q. CAPX and R&D coefficients and standard errors are multiplied by 100 (percentage points of the outcome ratio). Standard errors clustered by firm in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10: Investment robustness to cost-of-capital and discount-rate controls.

Specification	Baseline full	Baseline CoC/DR sample	+ CoC	+ DR	Obs. full	Obs. matched
Panel A. Attention-to-the-future coefficient						
Yearly						
SIC2-year FE	0.294***	0.279***	0.278***	0.284***	61,457	29,966
	(0.035)	(0.040)	(0.040)	(0.040)		
Yearly						
Firm + SIC2-year FE	0.221***	0.199***	0.198***	0.194***	60,852	29,572
	(0.024)	(0.030)	(0.030)	(0.030)		
Quarterly						
SIC2-quarter FE	0.040***	0.036***	0.037***	0.037***	229,066	113,164
	(0.005)	(0.006)	(0.006)	(0.006)		
Quarterly						
Firm + SIC2-quarter FE	0.022***	0.023***	0.023***	0.022***	228,810	113,033
	(0.003)	(0.004)	(0.004)	(0.004)		
Panel B. Added control coefficients						
Yearly						
SIC2-year FE			-0.494***	-0.257***		
			(0.066)	(0.043)		
Yearly						
Firm + SIC2-year FE			-0.140**	-0.343***		
			(0.062)	(0.033)		
Quarterly						
SIC2-quarter FE			-0.099***	-0.046***		
			(0.016)	(0.010)		
Quarterly						
Firm + SIC2-quarter FE			0.014	-0.038***		
			(0.014)	(0.007)		
Firm fixed effects	No for SIC2-time rows; yes for Firm + SIC2-time rows					
SIC2 × time fixed effects	Yes in all rows					
Log lag assets and Tobin's Q controls	Yes in all columns					

Notes: The table reports investment regressions corresponding to the selected size-only yearly and quarterly tables. The dependent variable is asinh(CAPX/lag assets). Attention to the future is standardized within each estimation sample and both investment and attention to the future are winsorized at the 1st and 99th percentiles. All columns control for log lag assets and Tobin's Q and cluster standard errors by firm. The yearly CoC/DR sample matches Gormsen-Huber data to the fiscal-year-end quarter; the quarterly sample matches exact quarters. Duplicate GVKEY-quarter CoC rows from multiple exchanges are averaged. Panel A coefficients and standard errors are multiplied by 100 and reported in percentage points of the investment ratio. Panel B reports the added control coefficient in percentage points of the investment ratio per 1 percentage point of the rate. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Financial-stress and uncertainty variables. The stress regressions in Table 3 use transparent indicators of distress and slack built from Compustat and market data—an operating loss (EBIT < 0), net leverage, the cash ratio, and a market-based distance-to-default—together with an earnings-miss indicator (reported earnings below the I/B/E/S analyst consensus), a stock-price-crash indicator (an annual return below -30%), and an overall-risk measure. The risk measure is a firm-level uncertainty index constructed from the same earnings-call transcripts following the text-based approach of Hassan et al. [2019] and provided by NL Analytics; because it shares its source with the attention measure, we enter it lagged one year to avoid mechanical overlap with the contemporaneous measure. Continuous stressors are standardized within the estimation sample (indicators enter per event), and the dependent variable is the year-on-year change in firm-year future attention.

B The Model: Derivations and Extensions

This appendix collects the derivations behind Section 3: the trading stage that underlies Lemma 1 (Appendix B.1), and the proofs of Propositions 1–3 (Appendices B.2–B.4). It then develops the ancillary engagement channel and the value-maximizing ownership structure that the main text states but does not derive (Appendix B.5). Notation is as in Section 3.

B.1 The trading stage and price informativeness

We use a signal-aggregation representation motivated by Edmans and Manso [2011], enriched so that the level of institutional ownership, and not only the number of owners, moves informativeness. Let v denote the innovation to the value the manager's action will create, Gaussian with prior precision normalized to one. Each institutional owner $i = 1, \ldots, I$ is informed and observes a private signal

$$\sigma_i = v + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}\big(0,\, 1/q(\beta)\big), \quad q'(\beta) > 0, \tag{11}$$

with ε_i independent across owners and of v. Signal precision $q(\beta) > 0$ is the intensive margin: we assume that a larger stock of institutional capital supports more information production, so each owner's signal is sharper. We use the share of posterior precision supplied by these signals as a reduced- form index of fundamental information passed into the price; we do not derive a noisy-order-flow trading equilibrium.

Proof of Lemma 1. The I conditionally independent signals carry total precision $I\, q(\beta)$ about v, which has prior precision one, so the posterior precision of v given the aggregate informed

signal is $1 + I\,q(\beta)$. We summarize price informativeness by the share of that posterior precision contributed by informed trading,

$$\Pi(\beta, I) = \frac{I\,q(\beta)}{1 + I\,q(\beta)} \in (0, 1). \tag{12}$$

We take Π to be the fidelity with which fundamentals pass into the price. A well-directed increment of reach shifts fundamental value by P, so the price moves by ΠP per unit and a manager holding μ of the equity captures $\mu\Pi P$ of the value her action creates. Differentiating,

$$\frac{\partial \Pi}{\partial \beta} = \frac{I\,q'(\beta)}{[1 + I\,q(\beta)]^2} > 0, \qquad \frac{\partial \Pi}{\partial I} = \frac{q(\beta)}{[1 + I\,q(\beta)]^2} > 0, \tag{13}$$

so the index rises with both the level and the breadth of institutional ownership. Setting $q \equiv 1$ collapses (12) to $\Pi = I/(I + 1)$, the Edmans and Manso [2011] benchmark in which only the number of owners moves informativeness. The microfoundation differs from theirs: in Edmans and Manso a single blockholder is already perfectly informed and the number of owners matters through competition in trading rather than through the arrival of additional signals. We adopt their reduced form for Π and obtain it by aggregation, which is what allows the level of institutional ownership to enter alongside the number of owners. □

B.2 Control and the endogenous blockholder

Proof of Proposition 1. Conditional on paying K and with the cap slack, the block's problem (3) is strictly concave in E, with interior solution $E(s) = s\,P(1 - \lambda)\,x$. Substituting back, the net value of controlling is

$$\Psi(s) = \tfrac{1}{2}\,s^2 P^2 (1 - \lambda)^2 x^2 - K, \tag{14}$$

strictly increasing in s for $x > 0$. Because overruling the manager once implements the value-maximizing action, a second intervening owner would duplicate the fixed cost K without creating additional value; control is therefore a single-owner activity, and the owner who undertakes it is the one with the highest net value Ψ. By monotonicity that is the largest block, m, and every smaller owner free-rides—the Shleifer and Vishny [1986] logic. Comparing entry with the zero-payoff outside option, the block participates iff $\Psi(m) \geq 0$, i.e. $\frac{1}{2} m^2 P^2 (1 - \lambda)^2 x^2 \geq K$, which rearranges to the threshold $m\,P(1 - \lambda)\,x \geq \sqrt{2K}$. When it holds the equilibrium threat is $E = m\,P(1 - \lambda)\,x$; when it fails no owner controls and $E = 0$. □

B.3 The manager's problem and equilibrium

Proof of Proposition 2. The manager's objective (5) is strictly concave in x, with first-order condition

$$x = \mu\Pi P\left[\lambda + (1-\lambda)E\right] + b\left[1 - (1-\lambda)E\right] = (\mu\Pi P\lambda + b) + (1-\lambda)(\mu\Pi P - b)\,E.$$

Substituting the block's interior rule $E = mP(1-\lambda)x$ from Proposition 1 gives

$$x = (\mu\Pi P\lambda + b) + mP(1-\lambda)^2(\mu\Pi P - b)\,x,$$

and solving for x, with the denominator positive by Assumption 1,

$$x^* = \frac{\mu\Pi P\lambda + b}{1 - mP(1-\lambda)^2(\mu\Pi P - b)}, \qquad E^* = mP(1-\lambda)\,x^*. \tag{15}$$

Setting $b = 0$ and, separately, $\Pi = 0$ (or $\mu = 0$) yields the two special cases in the statement. Finally, intervention is a probability, so $E \leq 1$. The unconstrained threat reaches the cap when $E^* = mP(1-\lambda)x^* \geq 1$, i.e. when $mP(1-\lambda)(\mu\Pi P + \lambda b) \geq 1$; control then saturates at $E^* = 1$, and evaluating the manager's first-order condition at $E = 1$ gives the finite corner action

$$x^* = \mu\Pi P + b\lambda \qquad (\text{corner, } E^* = 1). \tag{16}$$

This condition can hold in either regime; its role differs across them. In the equity-dominant region $\mu\Pi P > b$, where the denominator of (15) could otherwise fall to zero, the cap is reached strictly before it does so—$E^* = 1$ at $m = [P(1-\lambda)(\mu\Pi P + \lambda b)]^{-1}$, whereas $D = 0$ only at the larger $m = [P(1-\lambda)^2(\mu\Pi P - b)]^{-1}$, since $\mu\Pi P + \lambda b > (1-\lambda)(\mu\Pi P - b)$—so x^* never diverges. In the private-benefit region $b > \mu\Pi P$ the denominator already exceeds one and x^* is finite regardless; the cap, if it binds, merely truncates a large intervention threat. □

B.4 Comparative statics

Proof of Proposition 3. Write $A = \mu\Pi P\lambda + b > 0$ and $D = 1 - mP(1-\lambda)^2(\mu\Pi P - b) > 0$, so $x^* = A/D$.

(i) Level and breadth. Both A and D depend on Π, with $\partial A/\partial\Pi = \mu P\lambda$ and $\partial D/\partial\Pi = -\mu mP^2(1-\lambda)^2$. Hence

$$\frac{\partial x^*}{\partial \Pi} = \frac{\mu P\left[\lambda D + A\,mP(1-\lambda)^2\right]}{D^2} > 0, \tag{17}$$

every term being positive under $D > 0$. By the chain rule and Lemma 1, $\partial x^*/\partial\beta = (\partial x^*/\partial\Pi)(\partial\Pi/\partial\beta) > 0$ and $\partial x^*/\partial I = (\partial x^*/\partial\Pi)(\partial\Pi/\partial I) > 0$.

(ii) The block. A is independent of m, and $\partial D/\partial m = -P(1-\lambda)^2(\mu\Pi P - b)$, so

$$\frac{\partial x^*}{\partial m} = -\frac{A}{D^2}\frac{\partial D}{\partial m} = \frac{A\,P(1-\lambda)^2(\mu\Pi P - b)}{D^2}. \tag{18}$$

Since $A > 0$, $D^2 > 0$, and $P(1-\lambda)^2 > 0$, the sign is that of $\mu\Pi P - b$: the block deters the action when $b > \mu\Pi P$ and encourages it when $\mu\Pi P > b$.

(iii) The interaction. The factor $\mu\Pi P - b$ in (18) is strictly increasing in μ (at rate ΠP), so raising μ moves $\partial x^*/\partial m$ from the deterrence region toward the encouragement region, crossing zero at $\mu\Pi P = b$. Evaluating the cross-partial at that boundary, where $D = 1$ and $A = (1+\lambda)b$,

$$\left.\frac{\partial^2 x^*}{\partial m\,\partial\mu}\right|_{\mu\Pi P = b} = (1+\lambda)(1-\lambda)^2\, b\,\Pi P^2 > 0. \tag{19}$$

The interaction is thus strictly signed at the sign-change boundary. The prediction is local in this sense: far inside the private-benefit region the magnitude of the interaction can shrink or change sign, so the tested claim is that a blockholder's deterrent weakens as managerial equity rises toward the boundary, not that the cross-partial is signed globally. □

B.5 Engagement, firm value, and the optimal ownership structure

Engagement. Informed owners possess a second governance lever, the productive engagement (voice) of Edmans and Manso [2011]. Owner i chooses engagement effort $z_i \geq 0$ to maximize $s_i\,\phi\,\log(1+Z) - z_i^2/(2s_i)$, where $Z = \sum_j z_j$ is aggregate engagement and $\phi > 0$ scales its productivity: the benefit is the owner's pro rata share of a concave firm-value gain, and the cost is convex and stake-scaled, so a larger owner both values engagement more and spreads its cost over a larger governance apparatus. The first-order condition is $z_i = s_i^2\phi/(1+Z)$; summing over owners and using $\Omega = \sum_i s_i^2$ gives $Z(1+Z) = \phi\,\Omega$, hence

$$Z^*(\Omega) = \frac{-1 + \sqrt{1 + 4\phi\,\Omega}}{2}, \tag{20}$$

which is strictly increasing in the voice-concentration index Ω. Unlike control, engagement is interior for any $\Omega > 0$: it carries no fixed cost and is not monopolized. Holding total institutional ownership β fixed, dispersing ownership lowers Ω and hence Z^*; in the symmetric benchmark $s_i = \beta/I$ we have $\Omega = \beta^2/I$, so more owners weaken engagement even as they sharpen the price.

Firm value. Expected firm value sums the value created by the managerial action, net of whether the manager's project is implemented or overruled, and the value created by engagement:

$$V = P\,x\,[\lambda + (1-\lambda)E] + \phi\,\log(1+Z). \tag{21}$$

Engagement enters V additively and does not appear in the manager's problem (5); this is why it moves firm value but yields no prediction for attention to the future, and why the main text can characterize x^* without it.

The optimal ownership structure. The value-maximizing structure trades off three levers. Dispersing the informed fringe raises Π (improving price discipline and x^*) but lowers Ω (worsening engagement), the Edmans and Manso [2011] trading-versus-voice tension. Under parameters for which the two marginal effects cross, the optimal number of institutional owners is interior; otherwise the optimum is a boundary. The block size carries the intervention tension. Holding Π and Ω fixed and working in the empirically relevant private-benefit region $b > \mu\Pi P$, write $A = \mu\Pi P\lambda + b$ and $D = 1 + mP(1-\lambda)^2(b - \mu\Pi P) > 1$, so $x^* = A/D$ and $E^* = mP(1-\lambda)x^*$. The managerial-action component of firm value is

$$V_x \equiv P\,x^*[\lambda + (1-\lambda)E^*] = \frac{P\lambda A}{D} + \frac{mP^2(1-\lambda)^2A^2}{D^2}. \tag{22}$$

The first term is the value of the manager's own initiative, which the block deters ($x^* = A/D$ falls in m); the second is the value the block creates by overriding low-value projects, which the block raises. Differentiating, $\partial V_x/\partial m > 0$ at $m = 0$ (the override gain, $\propto P^2(1-\lambda)^2A^2$, dominates the deterrence loss, $\propto P\lambda A \cdot P(1-\lambda)^2(b-\mu\Pi P)$, because $A > \lambda(b-\mu\Pi P)$) while $\partial V_x/\partial m < 0$ for m large; the numerator of $\partial V_x/\partial m$ is linear in m with a unique root, so V_x is single-peaked. The value-maximizing block is interior, $m^* > 0$, when that root falls within the feasible range of block sizes (recall $m \le \beta$), and a corner otherwise. Combined with the control threshold of Proposition 1, an interior peak yields an extensive margin: a blockholder is active only where the value-maximizing block clears the fixed cost of control, $m^*P(1-\lambda)x^* \ge \sqrt{2K}$; otherwise the firm is dispersed and blockholder-free.

Two implications follow. First, the block is distinct from total institutional ownership β and from the voice-concentration index Ω: β (with I) sets price precision, Ω sets the free-rider strength of engagement, and m sets control—a separation the symmetric benchmark $m = \beta/I$ cannot deliver. Second, the attention-maximizing and value-maximizing structures differ: attention to the future falls monotonically with the block throughout the private-benefit region, whereas firm value is single-peaked in it, so the ownership structure that maximizes value need not be the one that maximizes attention to the future.

B.6 Robustness of the block–equity interaction

The $m \times \mu$ interaction of Table 7 is robust to absorbing nonlinearity in the block and to residualizing insider ownership with respect to block size. Table 11 adds the squared block m^2 (column 2), adds m^2 and μ^2 (column 3), and replaces μ with its component orthogonal to m (column 4). The interaction is stable throughout, ranging from 1.1% to 1.6% of a within-firm standard deviation per unit of the product of standardized components and remaining significant.

	Dependent variable: attention to the future (within SD)			
	(1) Baseline	(2) $+m^2$	(3) $+m^2, \mu^2$	(4) $\mu \perp m$
Blockholder m	-0.046^{***}	-0.035^{*}	-0.035^{*}	-0.036^{***}
	(0.014)	(0.019)	(0.019)	(0.012)
Insider ownership μ	-0.022	-0.025	-0.029	-0.020
	(0.017)	(0.018)	(0.023)	(0.015)
$m \times \mu$	0.014^{**}	0.016^{**}	0.015^{**}	0.011^{**}
	(0.006)	(0.007)	(0.008)	(0.006)
m^2		-0.004	-0.004	
		(0.005)	(0.005)	
μ^2			0.002	
			(0.008)	
β, breadth N, lagged assets	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Sector × year FE	Yes	Yes	Yes	Yes
Observations	35,416	35,416	35,416	35,416
Firms	3,593	3,593	3,593	3,593

Notes: Robustness of the block–equity interaction of Table 7. Column (1) repeats the interaction specification; (2) adds the squared block m^2; (3) adds m^2 and μ^2; (4) replaces μ with its component orthogonal to m ($\mu \perp m$). All columns include β, breadth N, a lagged-book-assets control, and firm and sector×year fixed effects; ownership standardized within sample. Coefficients are expressed in within-firm standard deviations ("within SD") of attention to the future (one within-firm SD ≈ 1.35 months; see Table 4): main-effect coefficients per one-SD increase in the regressor, interaction coefficients per one-unit increase in the product of standardized components. SE clustered by firm. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 11: Robustness of the conditional blockholder effect ($m \times \mu$).

C Ownership Structure: Variable Definitions and Robustness

The robustness tables—Tables 12 and 13 (institutional ownership) and Table 14 (ownership concentration)—examine a broader set of ownership-structure measures than the three used in the main text. All are constructed from the owner–firm–quarter positions in the Ownership Project data and averaged over the calendar year, and all holdings are expressed as a fraction of shares outstanding. The baseline 13F share is not top-coded and can exceed one when filing dates and the shares-outstanding denominator do not align. The broader total institutional ownership measure used in the robustness tables is top-coded at one. That cap creates a mass point: the broader measure has a 90th percentile of 1.000 (mean 0.85, median 0.93), so

the most heavily held firm-years are collapsed onto a single value—if anything attenuating its level coefficient toward zero. The names below match the row and column labels in the tables. The baseline level measure in Table 5 is *13F institutional ownership*, the summed reported holdings of 13F investment managers relative to shares outstanding. The broader robustness measure, *total institutional share ownership*, is the summed share of all 13F, 13D, and 13G filers, and *lagged total institutional share ownership* is its one-year lag. The filing-source rows split holdings into mutually exclusive components: *13F institutional ownership* (institutional investment managers), *13D active blockholder ownership* (control-intent 5% holders), *13G passive/exempt blockholder ownership* (passive or exempt 5% holders), and *F345 insider ownership* (officers, directors, and 10% owners filing Forms 3/4/5); positions are assigned first to 13D, then 13G, then 13F. *Vote ownership minus share ownership* is total votes controlled minus total shares held, capturing divergence between voting and cash-flow rights. *Largest institutional-owner share* is the holding of the single largest institutional owner in the 13F/13D/13G universe; *top owner minus second owner* is the gap between the largest and second-largest owners; and *owner HHI* is the Herfindahl index of ownership shares (the sum of squared owner shares). *Any owner above 20 percent* is the fraction of the year's quarters in which at least one owner holds more than 20% of shares outstanding (defined analogously at the 5% and 10% thresholds), and *ownership held by owners above 10 (20) percent* is the summed share of owners above that threshold. The *13D-or-13G blockholder share of institutional ownership* is the fraction of total institutional ownership reported through a 13D or 13G (>5%) blockholder disclosure. Full construction details, including the exact normalization, are given in the notes to Tables 12–14.

C.1 Robustness tables

Table 12: Alternative measures and dynamics of institutional ownership.

	Dependent variable: attention to the future (within SD)					
Panel A. Alternative ownership measures and dynamics						
	Total		Voting		Current + lagged	
	(1) SIC2-year FE	(2) Firm + SIC2-year FE	(3) SIC2-year FE	(4) Firm + SIC2-year FE	(5) SIC2-year FE	(6) Firm + SIC2-year FE
Total institutional share ownership	0.090***	0.039***	0.089***	0.039***	0.098***	0.030*
	(0.018)	(0.015)	(0.018)	(0.015)	(0.019)	(0.018)
Vote ownership minus share ownership			0.028*	-0.009		
			(0.017)	(0.021)		
Lagged total institutional share ownership					-0.005	0.001
					(0.020)	(0.017)
Log lag assets control	Yes	Yes	Yes	Yes	Yes	Yes
Controls excluding Tobin's Q	No	No	No	No	No	No
Firm fixed effects	No	Yes	No	Yes	No	Yes
SIC2 × year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	35,739	35,416	35,739	35,416	34,209	33,862
Firms	3,916	3,593	3,916	3,593	3,800	3,453
Within R^2	0.0219	0.0021	0.0222	0.0021	0.0237	0.0023
Mean dependent variable (months)	4.93	4.93	4.93	4.93	4.95	4.94

Notes: The table reports OLS regressions of attention to the future on institutional ownership. Coefficients and standard errors are expressed in within-firm standard deviations ("within SD") of the dependent variable (one within-firm SD ≈ 1.35 months; see Table 4). Listed ownership variables are standardized within the estimation sample, so coefficients correspond to a one-standard-deviation increase. Standard errors are clustered by firm and reported in parentheses. The Total columns include total institutional share ownership. The Voting columns add the total vote-share ownership wedge. The Current + lagged columns include current and lagged total institutional share ownership jointly. Total institutional ownership is computed from 13F, 13D, and 13G rows; F345-only rows are excluded. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 13: Institutional-ownership controls and filing-source decomposition.

	Dependent variable: attention to the future (within SD)			
Panel B. Controls and filing-source decomposition				
	Controls		Filing source	
	(7) SIC2-year FE	(8) Firm + SIC2-year FE	(9) SIC2-year FE	(10) Firm + SIC2-year FE
Total institutional share ownership	0.074***	0.036**		
	(0.020)	(0.016)		
13F institutional ownership			0.117***	0.085***
			(0.015)	(0.015)
13D active blockholder ownership			-0.027*	-0.030**
			(0.014)	(0.014)
13G passive/exempt blockholder ownership			-0.007	-0.021
			(0.015)	(0.013)
F345 insider ownership			-0.003	-0.003
			(0.017)	(0.017)
Log lag assets control	Yes	Yes	Yes	Yes
Controls excluding Tobin's Q	Yes	Yes	No	No
Firm fixed effects	No	Yes	No	Yes
SIC2 × year fixed effects	Yes	Yes	Yes	Yes
Observations	30,994	30,700	35,739	35,416
Firms	3,425	3,133	3,916	3,593
Within R^2	0.0328	0.0041	0.0243	0.0036
Mean dependent variable (months)	5.01	5.01	4.93	4.93

Notes: The table reports OLS regressions of attention to the future on institutional ownership. Coefficients and standard errors are expressed in within-firm standard deviations ("within SD") of the dependent variable (one within-firm SD ≈ 1.35 months; see Table 4). Listed ownership variables are standardized within the estimation sample, so coefficients correspond to a one-standard-deviation increase. Standard errors are clustered by firm and reported in parentheses. The Controls columns add lagged log sales, book leverage, operating cash flow/assets, and the cash ratio to the log lag assets control, excluding Tobin's Q. The Filing source columns enter 13F, F345, 13D, and 13G ownership shares jointly. 13D identifies active/control-intent blockholders, 13G identifies passive or exempt blockholder reporting, and F345 identifies insider Forms 3/4/5. Total institutional ownership is computed from 13F, 13D, and 13G rows; F345-only rows are excluded from that measure. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 14: Ownership concentration and blockholders.

	Dependent variable: attention to the future (within SD)						
Panel B. Ownership concentration and blockholders							
	Concentration			Blockholders			
	(1) Largest owner	(2) Top-owner gap	(3) Owner HHI	(4) Any owner >20%	(5) Ownership by >10% owners	(6) Ownership by >20% owners	(7) By filing status
Total institutional ownership	0.067*** (0.016)	0.050*** (0.015)	0.071*** (0.016)	0.052*** (0.015)	0.063*** (0.016)	0.059*** (0.016)	0.058*** (0.015)
Largest-owner share	-0.066*** (0.012)						
Top owner minus second owner		-0.048*** (0.011)					
Owner HHI			-0.067*** (0.012)				
Any owner above 20 percent				-0.038*** (0.011)			
Ownership held by owners above 10 percent					-0.047*** (0.012)		
Ownership held by owners above 20 percent						-0.054*** (0.012)	
13D-or-13G blockholder share of institutional ownership							-0.074*** (0.014)
Log lag assets control	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls excluding Tobin's Q	No	No	No	No	No	No	No
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SIC2 × year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	35,416	35,416	35,416	35,416	35,416	35,416	35,416
Firms	3,593	3,593	3,593	3,593	3,593	3,593	3,593
Within R^2	0.0035	0.0029	0.0035	0.0027	0.0028	0.0030	0.0035
Mean dependent variable (months)	4.93	4.93	4.93	4.93	4.93	4.93	4.93

Notes: The table reports OLS regressions of attention to the future on institutional ownership structure. Coefficients and standard errors are expressed in within-firm standard deviations ("within SD") of the dependent variable (one within-firm SD ≈ 1.35 months; see Table 4). Listed ownership variables are standardized within the estimation sample, so coefficients correspond to a one-standard-deviation increase. Columns (1)–(6) include total institutional share ownership and one concentration or blockholder measure; these coefficients therefore compare firms with similar total institutional ownership but different ownership concentration. All columns include log lag assets, firm fixed effects, and SIC2 × year fixed effects. Blockholder indicators are annual shares of quarters in which at least one institutional owner holds more than the listed share of shares outstanding. Ownership by blockholders is the annual mean normalized share held by institutional owners above the listed threshold. Column (7) splits blockholders by filing status: it includes total institutional share ownership and a single combined 13D-or-13G blockholder share of total institutional ownership. Ownership is assigned first to 13D, then to 13G, then to 13F, so 13F is the omitted filing-source category. Total institutional ownership is computed from 13F, 13D, and 13G rows; F345-only rows are excluded. Standard errors are clustered by firm and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D Russell IV: Full Specification

Table 15: Russell index-assignment IV: Passive ownership and attention to the future.

Panel A: event-time profile *(instrument dated t; months per +1pp of passive ownership)*	$t-1$ *(pre)*	t	$t+1$	$t+2$	
Passive ownership (2SLS)	-0.002	+0.040	+0.171***	+0.080	
	(0.056)	(0.057)	(0.057)	(0.053)	
AR 95% set	[-0.111, +0.109]	[-0.071, +0.155]	[+0.065, +0.292]	[-0.022, +0.189]	
Reduced form	-0.002	+0.042	+0.181***	+0.090	
First-stage F	99	100	103	115	
Observations	10,069	10,564	10,090	9,602	
Panel B: the $t+1$ estimate across specifications *(2SLS b, p house/reghdfe, AR, F, N)*					
Baseline: broad passive, ±500, firm+SIC2×yr	+0.171*** (0.057)	0.003 / 0.004	[+0.065, +0.292]	103	10,090
FE: firm + year	+0.179*** (0.064)	0.005 / 0.005	[+0.062, +0.317]	86	10,195
Window ±250	+0.265** (0.110)	0.016 / 0.025	[+0.070, +0.529]	35	4,793
Margin: Russell-benchmarked passive	+0.238*** (0.076)	0.002 / 0.003	[+0.092, +0.389]	545	10,090
Margin: ex-iShares passive (no same-source)	+0.454** (0.183)	0.013 / 0.017	[+0.164, +1.013]	17	10,090
+ size controls (assets, sales)	+0.167*** (0.053)	0.002 / 0.003	[+0.068, +0.278]	116	10,040
Cluster by gvkey	+0.171*** (0.057)	0.003 / 0.009	[+0.065, +0.292]	103	10,090
Years 2002–2020 (full FL sample)	+0.129** (0.054)	0.018 / 0.023	[+0.026, +0.241]	114	13,010

Notes: 2SLS of attention to the future (months, winsorized 1/99) on the broad S12 index/passive ownership share (percentage points, winsorized 1/99), instrumented by banding-aware predicted Russell 2000 assignment. Baseline: firm and SIC2×year fixed effects, quadratic ln(company cap), ±500-rank window around the Russell 1000/2000 cutoff, banding era 2007–2020, standard errors clustered by firm (permco). p is reported under the house dof convention and (rg) the reghdfe convention; AR is the weak-instrument-robust Anderson–Rubin 95% set. The estimand is the index/passive-ownership margin: the instrument does not move total institutional ownership ($F \approx 3$–5) and reduces the number of institutional owners, so the table is not evidence about total institutional ownership. The sample ends in fiscal 2020 to match the corporate-ownership data coverage used throughout the ownership tables. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 16 collects two reconstitution diagnostics. *Panel A* reports density diagnostics. The mechanical instrument panel is symmetric; in the headline estimation sample, pooled counts differ modestly across sides, but the estimated discontinuity at the cutoff is statistically insignificant. *Panel B* shows that the $t+1$ estimate remains positive across windows, fixed-effect menus, size controls, and cluster levels. Together with the statistically null pre-period estimate reported in Section 5.3, these results reinforce the temporal ordering behind the direction-of-causality test.

Table 16: Russell index-assignment IV: Reconstitution diagnostics.

Panel A: density of the running variable (counts per 50-rank bin, pooled 2007–2020)

Sample	N	Below/above	Symmetry z (p)	McCrary jump (SE)	t (p)
Instrument panel (mechanical)	14,014	0.500/0.500	+0.12 (0.906)	-2.4 (2.5)	-0.95 (0.356)
Headline estimation sample	10,090	0.518/0.482	+3.70 (0.000)	-11.1 (12.8)	-0.87 (0.399)

Panel B: sensitivity of the headline $t+1$ 2SLS estimate (months per +1pp of passive ownership)

Specification	2SLS b	(SE)	AR 95%	F	N
Baseline: ±500, firm+SIC2×yr, permco cluster	+0.171***	(0.057)	[+0.065, +0.292]	103	10,090
Window ±250	+0.265**	(0.110)	[+0.070, +0.529]	35	4,793
FE: firm + year	+0.179***	(0.064)	[+0.062, +0.317]	86	10,195
+ size controls (assets, sales)	+0.167***	(0.053)	[+0.068, +0.278]	116	10,040
Cluster by gvkey	+0.171***	(0.057)	[+0.065, +0.292]	103	10,090

Notes: Diagnostics for the banding-era Russell 1000/2000 IV (broad S12 passive share instrumented by banding-aware predicted R2000 assignment), ±500 cap-rank window, 2007–2020, attention-to-the-future sample, SEs clustered by firm (permco). Panel A: the running variable is the reconstructed end-of-May cap rank, assigned mechanically by Russell (one firm per rank per year), so the density check is pro forma and diagnoses sample coverage; the McCrary-style statistic is the jump in binned counts at the cutoff allowing separate linear trends on each side. Panel B re-estimates the headline $t+1$ 2SLS across windows, fixed-effect menus, size controls, and cluster levels; AR is the weak-instrument-robust Anderson–Rubin 95% set. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.